

MovementLogger — Sensor & Module Catalog

9 Angebote · 2026-05-19 15:35 UTC · BOM aus movement_logger_firmware · via crawl2pump

67 neu (7 Tage) · 0 aktualisiert · 0 touched · 0 price changes

Build & Wiring — LilyGO T-Beam S3 Supreme + XM125 radar

1 · Power / battery

- Insert one **charged 18650 Li-ion cell (flat-top, Ø18 × 65 mm)** into the LilyGO's onboard holder — mind polarity. Not included; order separately.
- USB-C charges the 18650 and runs the board via the onboard AXP2101 PMU. On the water run untethered off the 18650; recharge over USB-C between sessions.

2 · Radar wiring — Qwiic female-jumper (SparkFun colour code)

- Plug the **Qwiic end** of the female-jumper cable into the **XM125's Qwiic port**.
- Wire the 4 female leads to the LilyGO's **real I²C header (GPIO17/18)** — **NOT** the connector labelled "QWIIC socket" (that is UART1, GPIO43/44, and will not work):

Lead	XM125 (Qwiic)	→ LilyGO pin
Black	GND	GND
Red	3.3 V	3V3 (peripheral rail)
Blue	SDA	GPIO 17 (SDA)
Yellow	SCL	GPIO 18 (SCL)

XM125 default I²C address 0x52 — no clash with the onboard QMC6310 / BME280 / OLED, so it shares the bus fine. Zero soldering: Qwiic plug one end, push-on header pins the other.

3 · Firmware

- Firmware must **enable the AXP2101 peripheral/sensor power rail before any I²C access** (the LilyGO I²C bus hangs otherwise), init I²C on GPIO17/18, then drive the XM125 via the Acconeer A121 / SparkFun XM125 library.
- Stock Meshtastic does **not** read the radar — use custom firmware (movement_logger_firmware or an Arduino-ESP32 sketch).

4 · Enclosure

- Plastic (RF-transparent) waterproof box — the radar ranges **through** the wall, so the whole stack can be fully sealed. **No metal/carbon** case (blocks radar + GPS). If you keep the LPS22DF barometer, add a Gore-type vent membrane.

GPS / GNSS 1

LilyGO T-Beam S3 Supreme (GNSS+IMU+mag+baro+SD, USB-C)

OSS firmware

USB-C pluggable

☑ USB-C

☑ WiFi

☑ Bluetooth

☑ GPS

☑ Motion sensors

☑ SD-card

L×B×H 10 × 3.3 × 1.3 cm

MCU: ESP32-S3 · dual Xtensa LX7 @240 MHz · 8 MB flash / 512 KB SRAM (+PSRAM) · WiFi+BLE

OSS firmware: <https://github.com/meshtastic/firmware>

Buy (EU/CH): [Bastelgarage \(CH\)](#) · [ChipDepot \(CH\)](#) · [TinyTronics \(NL\)](#) · [OpenELAB \(EU\)](#) · [Elektor \(EU\)](#) · [The Machine Shop \(UK\)](#) · [Bot'n Roll \(PT\)](#) · [LilyGO \(direct\)](#)



neu LilyGO T-Beam S3 Supreme (GNSS+IMU+mag+baro+SD, USB-C) · T-Beam S3 Supreme @ LilyGO (vendor page)

USD 40.35

LilyGO (manufacturer store)

All-in-one, nothing to plug in: ESP32-S3 WiFi/BLE + QMI8658 6-axis IMU + QMC6310 mag + BME280 baro/temp/humidity + microSD slot + GNSS (u-blox MAX-M10 or Quectel L76K, variant-dependent) on one USB-C board. OSS ESP-IDF / Arduino-ESP32. ...

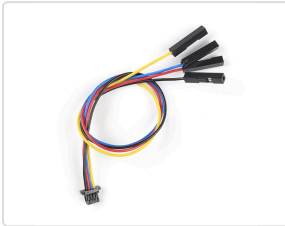
<https://www.lilygo.cc/products/t-beam-supreme-meshtastic>

SparkFun Flexible Qwiic Cable – Female Jumper (4-pin)

OSS firmware

Qwiic / I²C☐ USB-C☐ WiFi☐ Bluetooth☐ GPS☐ Motion sensors☐ SD-card

L×B×H 15 × 0.5 × 0.3 cm

**neu SparkFun Flexible Qwiic Cable – Female Jumper (4-pin) · CAB-17261 @ DigiKey**

CHF 1.52

SparkFun @ DigiKey · out of stock

Qwiic plug → 4 female Dupont leads. THE no-solder link for the LilyGO T-Beam S3 Supreme: its 'QWIIC socket' is UART1, so plug these onto its real I²C header — GPIO17 (SDA), GPIO18 (SCL), 3V3, GND. Connects the VL53L1X / XM125 to the LilyGO...

<https://www.digikey.com/en/products/detail/sparkfun-electronics/17261/13629019>
**neu SparkFun Flexible Qwiic Cable – Female Jumper (4-pin) · CAB-17261 @ Mouser**

CHF 1.55

SparkFun @ Mouser · in stock

Qwiic plug → 4 female Dupont leads. THE no-solder link for the LilyGO T-Beam S3 Supreme: its 'QWIIC socket' is UART1, so plug these onto its real I²C header — GPIO17 (SDA), GPIO18 (SCL), 3V3, GND. Connects the VL53L1X / XM125 to the LilyGO...

<https://www.mouser.ch/ProductDetail/SparkFun/CAB-17261?qs=DRkmTr78QAQLJE%2FDhtP97Q%3D%3D>
**neu SparkFun Flexible Qwiic Cable – Female Jumper (4-pin) · SparkFun 17261 @ sparkfun.com**

USD 1.95

SparkFun

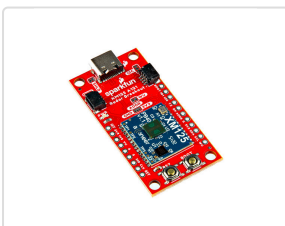
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<https://www.sparkfun.com/products/17261>
SparkFun XM125 60 GHz Radar (Acconeer A121, seal-in-box distance)

OSS firmware

Qwiic / I²C☐ USB-C☐ WiFi☐ Bluetooth☐ GPS☐ Motion sensors☐ SD-card

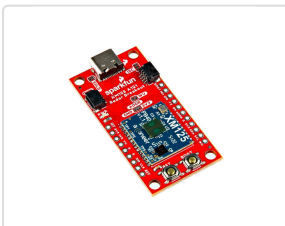
L×B×H 5.08 × 2.54 × 0.5 cm

OSS firmware: https://github.com/zdavatz/movement_logger_firmware**neu SparkFun XM125 60 GHz Radar (Acconeer A121, seal-in-box distance) · SEN-24540 @ DigiKey**

CHF 38.91

SparkFun @ DigiKey · in stock

60 GHz pulsed-coherent radar (Acconeer A121). Ranges through a sealed non-metal enclosure wall — no optical window needed, fully pottable in a plastic box. I²C/Qwiic: connect to the LilyGO's real I²C header (GPIO17 SDA / GPIO18 SCL / 3V3 / ...

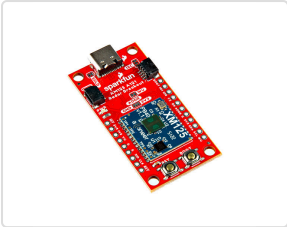
<https://www.digikey.com/en/products/detail/sparkfun-electronics/24540/22606868>
**neu SparkFun XM125 60 GHz Radar (Acconeer A121, seal-in-box distance) · SparkFun 24540 @ sparkfun.com**

USD 49.95

SparkFun

60 GHz pulsed-coherent radar (Acconeer A121). Ranges through a sealed non-metal enclosure wall — no optical window needed, fully pottable in a plastic box. Qwiic, plugs into the LilyGO T-Beam S3 Supreme's QWIIC socket / STEVAL I²C with the...

<https://www.sparkfun.com/products/24540>



neu SparkFun XM125 60 GHz Radar (Acconeer A121, seal-in-box distance) · SEN-24540 @ Mouser

SparkFun @ Mouser

60 GHz pulsed-coherent radar (Acconeer A121). Ranges through a sealed non-metal enclosure wall — no optical window needed, fully pottable in a plastic box. Qwiic, plugs into the LilyGO T-Beam S3 Supreme's QWIIC socket / STEVAL I²C with the...
<https://www.mouser.ch/ProductDetail/SparkFun/SEN-24540?qs=sqEgtWRSIJ2IBYifKTCuiw%3D%3D>

Battery / Power 2

18650 Li-ion cell — flat-top Ø18 × 65 mm (LilyGO power)

OSS firmware

18650 cell

- ☐ USB-C
- ☐ WiFi
- ☐ Bluetooth
- ☐ GPS
- ☐ Motion sensors
- ☐ SD-card
- L×B×H 6.5 × 1.8 × 1.8 cm

Buy (EU/CH): [nkon.nl](#) (EU 18650 specialist) · [Bastelgarage](#) (CH)



neu 18650 Li-ion cell — flat-top Ø18 × 65 mm (LilyGO power) · SparkFun 12895 @ sparkfun.com

USD 9.83

SparkFun

NOT included with the LilyGO T-Beam S3 Supreme — buy separately. Must be FLAT-TOP, Ø18 × 65 mm (button-top cells are longer and won't seat). 3.6–3.7 V Li-ion, ≥2500 mAh recommended; the board's AXP2101 PMU charges it over USB-C and runs...
<https://www.sparkfun.com/products/12895>



neu 18650 Li-ion cell — flat-top Ø18 × 65 mm (LilyGO power) · Generic 18650 @ Generic 18650 (vendor page (reference))

Generic 18650 (manufacturer store)

NOT included with the LilyGO T-Beam S3 Supreme — buy separately. Must be FLAT-TOP, Ø18 × 65 mm (button-top cells are longer and won't seat). 3.6–3.7 V Li-ion, ≥2500 mAh recommended; the board's AXP2101 PMU charges it over USB-C and runs...
<https://www.nkon.nl/rechargeable/18650-size.html>